

Will Ruff

425-615-0778 | wruff9150@sdsu.edu | [linkedin.com/in/willhruff](https://www.linkedin.com/in/willhruff) | github.com/willruff

EDUCATION

San Diego State University

Bachelor of Science in Computer Science, Minor in Mathematics

Major GPA: 4.0 | Cumulative GPA: 3.86

San Diego, CA

Aug. 2022 – May 2026

RELEVANT COURSEWORK

- Data Structures
- Database Theory Implementation
- Object Oriented Programming
- Adv. Programming Languages
- Operating Systems
- Computer Architecture

EXPERIENCE

Operating Systems Teaching Assistant

Jan 2025 – Present

San Diego State University

San Diego, CA

- Assisted in maintaining and configuring the auto-grader codebase to ensure accurate and efficient grading
- Set up and tested the auto grader for each assignment, optimizing performance and creating unique test cases
- Graded assignments and exams for a class of 150 students ensuring consistency and adherence to grading rubrics
- Held weekly office hours to provide personalized support, debugging assistance, and address student questions
- Collaborated with the instructor to improve course materials and enhance the students' learning experience

Software Engineer Intern

June 2024 – Jan. 2025

Glympse - Location Sharing Technology

Seattle, WA

- Developed Python and Go scripts for file comparison, cutting memory usage by 25% via benchmarking
- Migrated legacy code to a new repository, adding essential parameters and enhancing report functionality
- Refactored and modernized a shell script report in Python, enhancing readability and maintainability
- Built and implemented comprehensive unit tests to ensure the reliability and accuracy of the Python code
- Deployed code using GitLab CI/CD pipelines, configuring environments with '.gitlab-ci.yml' and Docker

Computer Science Research Assistant

Feb. 2024 – Dec. 2024

San Diego State University

San Diego, CA

- Investigated detailed medical device recalls to detect underlying problems related to software issues
- Implemented efficient code to automate data extraction of over 1000 medical device recalls from the FDA website
- Engaged in collaborative problem-solving exercises to address research challenges and advance objectives
- Utilized HTML5 to design a compelling and operational website that proficiently showcased research

PROJECTS

StockScript | Python, TextX

Nov. 2024 – Dec. 2024

- Designed and implemented a domain-specific programming language for stock and crypto modeling
- Developed grammar definitions using TextX and created a Python interpreter to execute scripts effectively
- Created user-friendly stock and crypto declarations with operations like price changes, merges, and splits
- Developed display features to present asset data, including prices, market caps, and supply metrics
- Added a simulation mode to generate good, bad, or random days for specific stocks or the entire market

Restaurant Flow Control | C++, Linux, VirtualBox

Nov. 2024

- Designed a restaurant simulation to manage seating and customer flow with a producer-consumer model
- Utilized a buffer to coordinate threads for seat request types, ensuring efficient resource allocation
- Used mutexes and semaphores to ensure thread safety and prevent deadlocks in multithreaded operations
- Critically analyzed resource utilization metrics to identify bottlenecks and improve overall system efficiency

Spam Detection AI Model | Python, Tensorflow, Keras, Pandas

Sep. 2023 – Dec. 2023

- Collaborated with a team of five to develop an AI model using TensorFlow and Keras to detect spam messages
- Utilized a dataset of over 10,000 messages, preprocessed extensively with Pandas, to train the model in Python
- Achieved a remarkable 98% accuracy in detecting spam messages through rigorous debugging and testing
- Created an engaging slide deck and confidently presented the final project to the SDSU Artificial Intelligence Club

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, HTML/CSS, TextX

Frameworks & Technologies: Git, Docker, CI/CD Pipelines, Linux, VirtualBox, VS Code, GNU Debugger, unittest